Image Fragment Reconstruction via Adjacency Prediction

Case Study Report

1. Model Architecture

The model is a **Siamese CNN** that operates on pairs of 16×16 fragments. A shared encoder maps each fragment to a 256-dimensional embedding; a single comparison head maps the two embeddings to a scalar confidence score $p_{ij} \in [0, 1]$.

Encoder. Three Conv–BatchNorm–ReLU blocks with max-pooling after the last two reduce the spatial dimensions from 16×16 to 4×4; a linear layer projects the flattened features to 256-d (ReLU + dropout $p = 0.3$).

Comparison head. Inspired by InferSent [?], the head takes the 1024-dimensional vector $[\mathbf{a} - \mathbf{b} \parallel \mathbf{a} \odot \mathbf{b} \parallel \mathbf{a} \parallel \mathbf{b}]$ as input, passes it through one hidden layer (ReLU), and produces a single scalar via sigmoid. Using the difference and Hadamard product alongside the raw embeddings gives the head explicit access to asymmetric relational features at no additional encoder cost.

Why Siamese? Weight sharing ensures the comparison is symmetric and forces the encoder to learn a universal fragment representation rather than one tied to a specific input slot. The pairwise architecture avoids the need for any image-level label: the only supervision required is whether two fragments are adjacent, which is derivable from the grid structure alone.

2. Training Strategy

Pretext task. The model is trained to predict spatial adjacency: $y_{ij} = 1$ iff fragments i and j are grid-neighbours within the same image. With $N = 10$ images on a 4×4 grid, the negative-to-positive ratio is exactly 12,480/240 = 52. We use weighted BCE with tilt parameter β:

$$\mathcal{L}_{\text{WBCE}}(\beta) = -\frac{1}{N} \sum_{(i,j)} \left[\beta{\cdot}52\, y_{ij} \log p_{ij} + (1-y_{ij}) \log(1-p_{ij})\right].$$

Multi-task loss. Adjacency is a local signal; the downstream task is global clustering. A same-image loss term $\mathcal{L}_{\text{BCE}}^{\text{same}}$ is added to bridge this gap, acting on the same score p_{ij}:

$$\mathcal{L} = \lambda_{\text{adj}}\, \mathcal{L}_{\text{WBCE}}(\beta) + \lambda_{\text{same}}\, \mathcal{L}_{\text{BCE}}^{\text{same}}.$$

Inference. The 160×160 matrix of scores p_{ij} is used directly as a similarity matrix for **balanced spectral clustering**: spectral embedding (normalised graph Laplacian, $k = 10$ components) followed by Hungarian-assignment balanced k-means that enforces exactly 16 fragments per cluster.

Training details. 1.28M training images (ImageNet-64); each step samples 10 fresh images. Up to 25,000 iterations; early stopping with patience 10, evaluated every 250 steps over 20 validation batches. All runs on CPU (Apple M-series).

3. Evaluation Results

Performance is measured by the **Adjusted Rand Index** (ARI), which counts pair-level agreement between predicted and true clusters, adjusted for chance. ARI = 0 is random; ARI = 1 is perfect.

Validation results. A five-seed study (seeds 0–4) across five hyperparameter configurations (Table **??**) shows ARI in the range 0.50–0.58, stable across seeds ($\sigma \approx 0.02$–0.03). A Welch one-way ANOVA finds no statistically significant difference between configurations ($\alpha = 0.10$), indicating the choice of loss weights is not a strong lever at this scale.

Configuration	**Mean ARI**	**SD**
(a) adj-only, $\beta = 1/52$	TODO	TODO
(b) same-only	TODO	TODO
(c) multi-task $\lambda_s = 1.0$	TODO	TODO
(d) multi-task $\lambda_s = 0.2$	TODO	TODO
(e) adj-only, $\beta = 1$	TODO	TODO

Table 1: Validation ARI (5 seeds each).

Test results. The best checkpoint (run `TODO`) was evaluated on 1,000 independent test batches: ARI = TODO ± TODO, NMI = TODO ± TODO, purity = TODO ± TODO, adjacency F1 = TODO.

Pretext task. Adjacency AUROC sits at 0.95–0.97 across all configurations, indicating the model has nearly solved the binary pretext task. The remaining gap to perfect clustering is attributed to fragments with near-uniform colour or repetitive texture that look similar across different source images.

4. Challenges and Improvements

Class imbalance. The 52:1 negative-to-positive ratio in adjacency pairs requires careful loss weighting. We found $\beta = 1/52$ (plain BCE) works as well as the textbook class-balanced $\beta = 1$; heavier upweighting degrades performance.

Local vs. global signal. Adjacency is a nearest-neighbour signal; two fragments from opposite corners of the same image are labelled negative. The same-image loss term partially closes this gap but does not produce a statistically significant improvement at this scale, suggesting the binding constraint is fragment ambiguity rather than the loss formulation.

Fragment ambiguity. Uniform-colour or repetitive-texture patches yield near-identical embeddings regardless of source, fundamentally limiting the achievable ARI for any pairwise model without global context.

5. Next Steps

Pretrained encoder. Replacing the trained-from-scratch CNN with a self-supervised encoder (e.g. DINOv2) would provide richer features at 16×16 resolution at no labelling cost, likely the highest-impact single change.

Topology-aware clustering. The true clustering is more constrained than "equal sizes": each cluster's induced subgraph must be isomorphic to a 4×4 grid. A GNN-based approach that exploits this planarity constraint could resolve borderline assignments that spectral clustering misses.

Larger-scale statistical study. With $n = 5$ seeds the ANOVA is underpowered for effect sizes below ~ 0.07 ARI. A 20-seed study would give $\Delta_{\min} \approx 0.03$, sufficient to detect the differences suggested by the single-seed sweep.

Color-histogram baseline. A non-learned baseline using per-fragment color histograms with spectral clustering would anchor the model's ARI against a trivial reference and quantify how much benefit comes from learned representation versus color matching.

References

[1] Conneau et al. (2017). *Supervised Learning of Universal Sentence Representations from Natural Language Inference Data*. EMNLP.